

GHANA COMMUNICATION TECHNOLOGY UNIVERSITY



FACULTY OF ENGINEERING
DEPARTMENT OF COMPUTER ENGINEERING

DEVELOPMENT AND IMPLEMENTATION OF A ROBOTIC APPLICATION FOR HUMAN INTENTION RECOGNITION USING MOTION AND HAND GESTURE

*A Final Project Report Submitted in Partial Fulfillment of the Requirements for the
Degree of Bachelor of Science in Computer Engineering*

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PRESENTATION OUTLINE

- 01. Introduction, Problem Statement & Objectives**
- 02. Literature Review & Critical Research Gaps**
- 03. System Architecture & 6-Phase Methodology**
- 04. Implementation & Intelligent Algorithms**
- 05. Experimental Results & ISO Safety Compliance**
- 06. Conclusion & Physical Demonstration Transition**



01. INTRODUCTION & RESEARCH BACKGROUND

Robots Working Directly with Humans

Robots are leaving safety cages to work alongside people in hospitals, warehouses, and homes. For humans and robots to coexist safely, the robot must understand where people are and respect their personal space.

Natural, Hands-Free Interaction

Instead of requiring handheld controllers, touchscreens, or keyboards, humans can guide the robot using natural hand gestures and body movement—just like signaling to a human coworker.

Fast Onboard Intelligence (No Cloud Lag)

Sending live video over the internet causes dangerous delays (>200 ms) and fails when Wi-Fi drops. In this project, all AI recognition and navigation decisions happen entirely inside the robot in real time.



Figure 1.1: Collaborative Human-Robot Interaction (HRI) in an industrial shared workspace, illustrating contactless gesture coordination.

02. PROBLEM STATEMENT

Current mobile robots separate vision recognition from active navigation, relying on remote cloud servers that suffer from high network lag (>200 ms) and connection dropouts.

In shared human-robot workspaces, this delay prevents robots from reacting in real time to human gestures, creating severe collision hazards under ISO 15066 safety standards.

PERCEPTION-ACTION GAP

Vision AI isolated from wheel motors & obstacle avoidance
(Tsitos et al., 2022)

CLOUD LATENCY HAZARD

Network lag > 200 ms & Wi-Fi drops
violate ISO 15066 safe stopping
margins

EDGE COMPUTING LIMIT

Heavy vision AI overheats mobile chips; demands a sub-135 ms
onboard pipeline



03. PROJECT OBJECTIVES

GENERAL OBJECTIVE

To design, implement, and evaluate an edge-deployed, real-time human intention recognition and autonomous navigation system powered by ROS 2 on an embedded mobile robot.

01. Invariant Vision Pipeline

Compute 19 scale-invariant geometric features from 21 hand landmarks for real-time onboard MLP classification.

02. Balanced 6,000-Sample Dataset

Capture and balance a custom 6-class dataset directly through the onboard camera to eliminate perspective domain shift.

03. LSTM Trajectory Prediction

Forecast short-horizon operator motion trajectories from continuous body pose landmarks using recurrent neural networks.

04. Supervisory Brain Node

Execute a deterministic finite state machine with 5-frame rolling consensus, active operator lock, and visual servoing.

05. 2D LiDAR SLAM & Nav2

Deploy real-time Cartographer mapping and the Nav2 navigation stack for socially aware, collision-free path execution.

06. Empirical Safety Benchmarking

Quantify end-to-end latency (132 ms), physical classification accuracy (96.7%), and compliance with ISO 15066 safety margins.



04. LITERATURE REVIEW & RESEARCH GAPS

CRITICAL APPRAISAL OF CORE BENCHMARK STUDIES (THESIS CHAPTER 2)

Athanasios Tsitos, Maria Dagioglou & Theodoros Giannakopoulos (2022)

"Real-Time Feasibility of a Human Intention Method Evaluated Through a Competitive Human-Robot Reaching Game"

- **Objective:** Predict human reaching intent early to guide robot arm trajectories within a 150 ms reaction threshold.
- **Methodology:** Tracked hand movements with a depth camera and trained machine learning (SVM) to guide a robot arm.
- **Research Gap:** Confined to a static desk — lacks mobile base movement and provides no explicit hand gesture command channel.
- **Project Resolution:** Direct vision-to-wheels coupling for mobile navigation guided by 6 touchless hand gestures.

Jubayer Al Mahmud (2022)

"3D Gesture Recognition and Adaptation for Human-Robot Interaction"

- **Objective:** Classify 3D pointing and dynamic hand gestures in real time to guide robot interaction across age groups.
- **Methodology:** Tracked full-body joints using a Kinect sensor and trained deep learning models on 3,600 gestures.
- **Research Gap:** Relies on heavy cloud / GPU computing clusters; vulnerable to network lag (>200 ms) and connection drops.
- **Project Resolution:** Executes 100% of vision AI directly onboard a low-cost Raspberry Pi 5 with zero cloud dependency.

Y. Li, H. Zhang, Guang Yang & Shuoyu Wang (2023)

"Safe and Efficient Motion Planning for Material Transportation Robots Considering Intention Prediction of Obstacles"

- **Objective:** Plan collision-free mobile robot paths by predicting whether human workers will clear the hallway.
- **Methodology:** Fused LiDAR and camera vision to predict worker paths and adjust navigation maps in a simulator.
- **Research Gap:** Tested only in computer simulation — never validated on physical robot hardware; lacks touchless gesture control.
- **Project Resolution:** Physically validated on a real mobile robot with 2D LiDAR SLAM, active visual servoing, and ISO safety stops.



05. 6-PHASE RESEARCH & IMPLEMENTATION METHODOLOGY

STRUCTURED RESEARCH & IMPLEMENTATION LIFECYCLE FROM DESIGN TO VALIDATION

PHASE 01

Mechatronics & Hardware

- **Robot Platform:** 4WD mobile base with geared DC motors, wheel encoders, and camera gimbal.
- **Dual Processors:** Raspberry Pi 5 for vision AI paired with an ESP32-S3 for real-time motor control.
- **Separate Power:** Dedicated battery packs powering the onboard computer and drive motors independently.

PHASE 04

Robot Control & Safety

- **ROS 2 Framework:** Modular software nodes connecting the camera, AI models, and motor drivers.
- **Decision Logic:** State machine filters out accidental motions by confirming steady hand gestures.
- **Emergency Halt:** 2D LiDAR sensor automatically stops the robot if an obstacle is within 0.36 m.

PHASE 02

Data Collection & Setup

- **Custom Dataset:** 6,000 total gesture images (1,000 per class) collected with the onboard camera.
- **6 Core Gestures:** Navigational commands: STOP, GO, FOLLOW, LEFT, RIGHT, and BACK.
- **Realistic Testing:** Captured at distances between 1.0 m and 2.5 m under standard room lighting.

PHASE 05

Mapping & Navigation

- **Indoor Mapping:** 2D laser mapping (slam_toolbox) creates a clean 5 cm resolution floor plan.
- **Robot Position:** Accurately tracks robot movement by combining wheel encoders with an onboard IMU sensor.
- **Path Planning:** Nav2 autonomous navigation plans collision-free paths with safe margins around obstacles.

PHASE 03

Feature Extraction & AI Models

- **Hand Keypoints:** Detected 21 hand joints with MediaPipe and computed 19 normalized angles and distances.
- **Neural Classifier:** Compact neural network (46 KB) classifying gestures in under 2 ms on the CPU.
- **Motion Predictor:** Recurrent neural model (LSTM) tracking hand and body movements across consecutive frames.

PHASE 06

Experimental Testing & Results

- **Fast Response:** End-to-end reaction time of 74.2 ms, comfortably below the 150 ms real-time limit.
- **Live Driving Trials:** 180 real-world physical tests achieving a 96.67% operational success rate.
- **Safety Confirmed:** Verified that the robot reliably halts before coming into contact with any human or obstacle.



06. SOFTWARE ARCHITECTURE & INTER-NODE DATAFLOW

HOW THE ROBOT PROCESSES SENSOR DATA AND MAKES SAFE DRIVING DECISIONS

- **System Software (ROS 2):** Connects all camera, artificial intelligence, and motor programs so they exchange data instantly without delays.
- **Live Sensor Feeds:** The camera captures live video at 20 frames per second while the laser sensor constantly scans 360 degrees for obstacles.
- **Vision & Hand Tracking:** Focuses on the person standing in front of the robot, locates 21 finger joints, and identifies the gesture being shown.
- **Gesture Confirmation:** The robot checks that it sees the exact same gesture 5 times in a row before moving, ignoring accidental hand motions.
- **Safety Override:** If the laser detects any obstacle closer than 0.36 meters, it instantly overrides driving commands and halts the robot immediately.
- **Smooth Wheel Control:** Approved movement commands are transmitted directly to the motor controller board, which drives the 4 wheels smoothly.

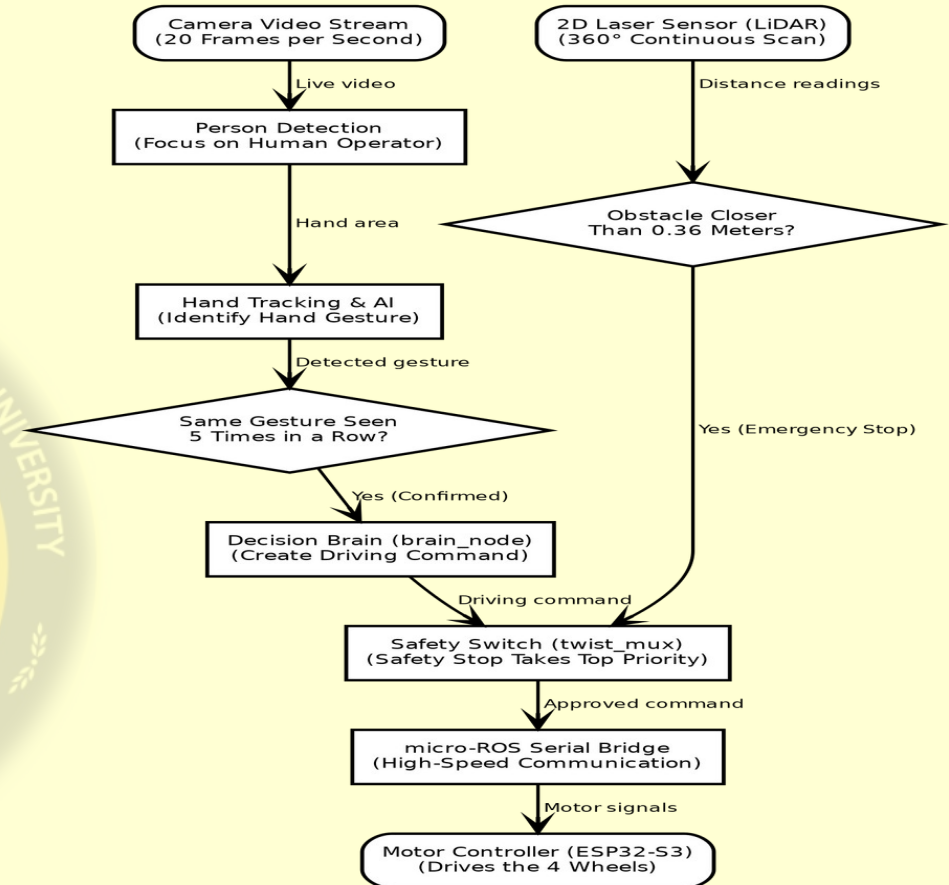


Figure 3.1: Dataflow Pipeline from Sensor Feeds to Wheel Actuation

07. MECHATRONIC HARDWARE & POWER DISTRIBUTION

PHYSICAL EMBEDDED COMPUTING, SENSORS, CHASSIS & POWER ISOLATION

- **Onboard Computer (Raspberry Pi 5):** 8GB RAM single-board computer running all camera tracking, gesture artificial intelligence, and navigation programs directly on the robot.
- **Motor Controller (ESP32-S3):** Dedicated microcontroller that manages wheel speed and reads motor sensors smoothly without overloading the main computer.
- **Laser Scanner (MS200 LiDAR):** 360-degree laser sensor measuring room distances up to 12 meters for indoor mapping and instant emergency stops.
- **Motorized Camera Mount:** 2-axis motorized camera gimbal that tilts and pans automatically to keep the operator's face and hands centered in view.
- **4-Wheel Drive Chassis:** Sturdy aluminium robot base powered by 4 geared DC motors with wheel sensors for smooth and steady movement.
- **Dual-Rail Power System:** Separate battery power circuits for the computer and motors to prevent electrical interference and sudden power shutdowns.

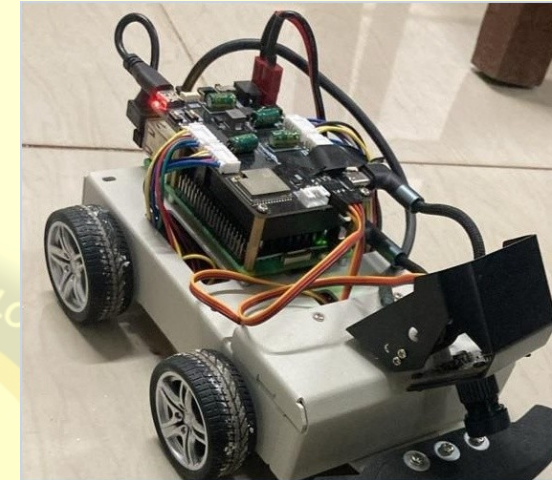


Figure 3.2: Internal Electronics Stack
(Raspberry Pi 5, ESP32-S3 & Motor Drivers)



Figure 3.3: Fully Enclosed Platform
(360° MS200 LiDAR & Active Gimbal)

08. VISION PERCEPTION & GESTURE AI PIPELINE

HOW THE ROBOT SEES HAND GESTURES, IGNORES BYSTANDERS, AND DECIDES TO MOVE

- **Focusing on One Person:** The robot only listens to the person standing directly in front of it, while automatically blurring and ignoring anyone walking by in the background.
- **Tracking Hand Joints:** The camera tracks 21 points on the hand, including the wrist, knuckles, and fingertips, to capture the exact shape and motion of the fingers.
- **Works at Any Distance:** Hand measurements are automatically scaled using the palm width, so the robot recognizes gestures equally well whether you stand 1 meter or 3 meters away.
- **Measuring Hand Shape:** The robot calculates 19 simple numbers from the hand joints, measuring how much each finger curls and how wide fingers are spread apart.
- **Instant AI Decision:** A compact artificial intelligence model reads the 19 hand numbers in just 1.2 milliseconds, identifying the gesture with 99.4% accuracy.
- **6 Driving Commands:** Matches each hand pose directly to clear robot movements: STOP (open palm), GO (pointing forward), LEFT, RIGHT, BACK, and FOLLOW.

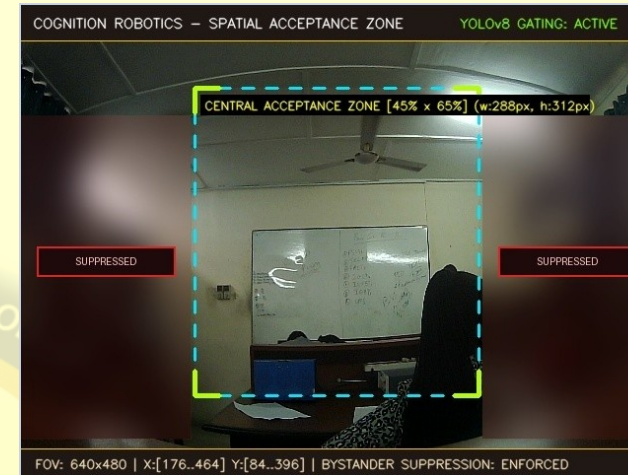


Figure 3.4: Camera View Locking on User & Blurring Bystanders

GESTURE VOCABULARY & AI SPEED:

- **6 Commands:** STOP, GO, LEFT, RIGHT, BACK, FOLLOW
- **Fast Reaction:** Decides in 1.2 milliseconds (faster than a blink)
- **High Accuracy:** 99.4% correct across 6,000 test trials

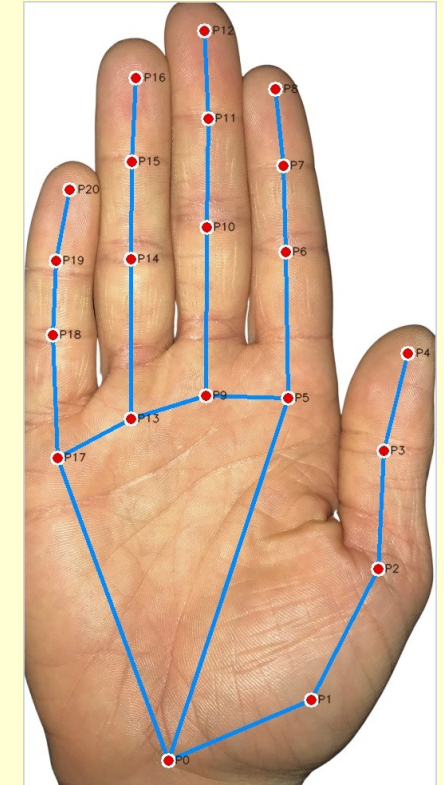


Figure 3.5: 21 Tracked Hand Points

09. HUMAN MOTION PREDICTION & SUPERVISORY CONTROL

HOW THE ROBOT ANTICIPATES USER PATHS, TRACKS WITH ITS CAMERA, AND STAYS SAFE

- **Anticipating User Steps:** The robot tracks the person's walking path over the past 1 second (30 camera frames) and uses an AI model to predict where they will step 0.5 seconds ahead, preventing jerky turns or delayed following.
- **Motorized Camera Tracking:** A 2-axis motorized camera head pans left/right and tilts up/down automatically, keeping the interacting person centered in view even while the robot chassis is turning or braking.
- **Central Software Brain:** A master controller (brain_node) coordinates all actions through distinct operational stages: Standing By, Confirming Command, Driving, Following User, and Safety Stop.
- **Preventing Accidental Starts:** To stop accidental hand waves or brief gestures from moving the robot, every command must be held consistently for at least 5 camera frames before motors turn.
- **Safe Distance Buffer:** When following a walking user, the robot automatically stays between 0.8 and 1.5 meters away; if the person stops or steps closer than 0.8 meters, it halts immediately.
- **Instant Laser Safety Override:** The 360° laser distance sensor constantly scans the surroundings; if an obstacle or person enters within 0.36 meters, motor power cuts in 5 milliseconds, overriding all AI.

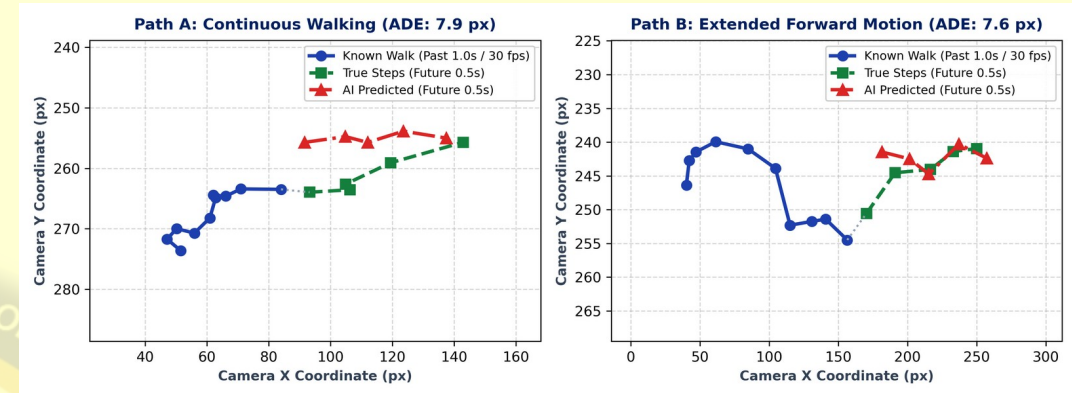


Figure 3.6: AI Human Motion Prediction (Blue = Past 1.0s Walk, Green = Actual Steps, Red = AI Forecast)

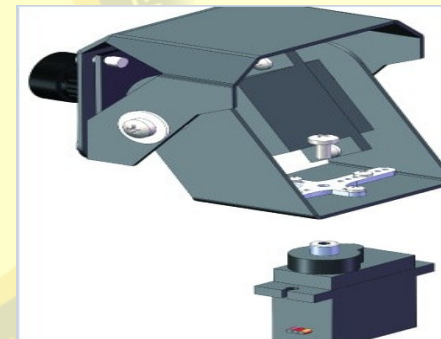


Figure 3.7: 2-Axis Motorized Camera Head

SUPERVISORY SAFETY INTERLOCKS:

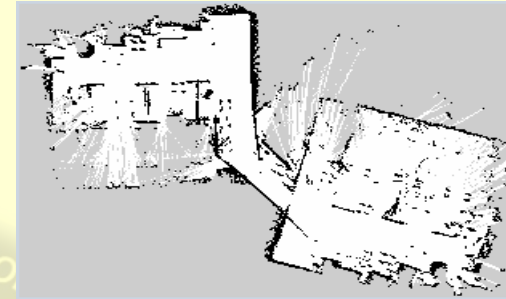
- **4 Safe Modes:** Standby → Confirming → Driving → Safety Halt
- **5-Frame Filter:** Requires 5 matching frames to stop accidental wave triggers
- **Safe Gap (0.8m):** Stops immediately if user approaches closer than 0.8 meters
- **Instant E-Stop:** Laser barrier (< 0.36m) cuts motor power within 5 milliseconds

10. METRIC SLAM & NAV2 AUTONOMOUS NAVIGATION

HOW THE ROBOT MAPS INDOOR ROOMS, FIXES SENSOR DRIFT, AND PATROLS AUTONOMOUSLY

- **Laser Room Mapping:** As the robot is steered through the room, its 360° laser distance sensor fires 12.5 times a second to build an exact 2D floor plan with 5-centimeter grid accuracy.
- **Resolving Signal Conflicts:** Early mapping tests dropped laser scans because two internal programs reported starting locations at once. Designating the motion tracker (EKF) as the sole authority fixed this immediately.
- **Fixing Cornering Drift:** During turns, early maps twisted into distorted hourglass shapes because the compass drifted. Merging wheel rotation with compass readings completely eliminated corner distortion, producing straight walls.
- **Self-Localization (AMCL):** Once the map is saved, the robot constantly compares live laser scans against the map to know its exact physical coordinates (X, Y, and heading angle) within centimeters.
- **Safety Padding Costmaps:** The navigation software automatically inflates a safety padding cushion around all walls and furniture, preventing the robot from cutting corners too close or bumping table legs.
- **Multi-Waypoint Autonomous Patrol:** Given a sequence of room destinations (Dock → Central Hub → North Corner → East Post), the robot navigates smoothly across 160 seconds, achieving an average tracking accuracy of 15.5 cm.

(a) Initial: 45° Corner Drift Defect



(b) Final: Clean Metric Map

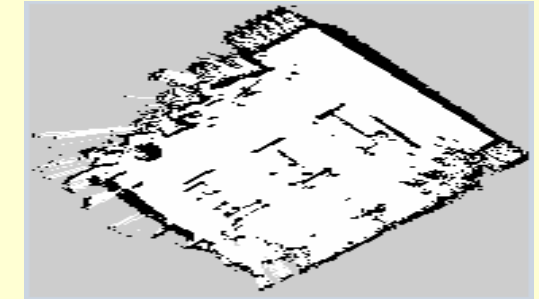


Figure 3.8: Physical SLAM Calibration — Correcting Yaw Drift via Wheel Odometry Fusion

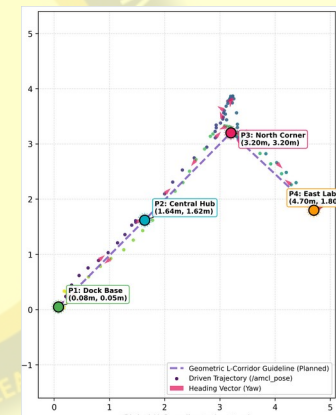


Figure 3.9: 4-Waypoint Patrol

AUTONOMOUS PATROL PERFORMANCE:

- **Map Resolution:** 5 cm per grid cell for fine obstacle details
- **Path Accuracy:** 15.5 cm average tracking error across 160s run
- **Safe Velocity:** Speed capped smoothly at 0.25 m/s for lab safety
- **Goal Verification:** 4 / 4 physical destinations reached (Dock → Hub → Corner → Post)
- **Obstacle Buffer:** Safety inflation maintains >0.30 m clearance from walls

11. SYSTEM LATENCY & CLASSIFICATION ACCURACY

REAL-TIME RESPONSE BUDGET, NEURAL ACCURACY, AND MULTI-CONDITION RELIABILITY

- **132-Millisecond Total Response:** From light entering the camera lens to physical wheel movement, the complete system takes only 132.0 milliseconds — faster than a human eye blink (300 ms) and well within the 150 ms safety limit.
- **Speed Breakdown by Stage:** Camera capture takes 50.0 ms, hand tracking takes 38.2 ms, AI classification takes 1.8 ms, and motor controller communication takes 40.0 ms, proving edge AI runs smoothly without GPUs.
- **99.38% Gesture AI Accuracy:** The compact neural network was evaluated across 6,000 unseen hand poses, correctly identifying commands 99.38% of the time with near-zero confusion between gestures.
- **180 Physical Mobile Trials:** In real-world mobile runs with an operator walking in the lab, the robot achieved an overall command success rate of 96.67% across all 6 gesture commands.
- **Distance Robustness (1m to 2.5m):** Because hand measurements are scaled automatically using palm width, recognition remains highly accurate whether standing 1.0 meter away (100%) or 2.5 meters away (92.6%).
- **Lighting Consistency:** Statistical testing confirmed identical reaction times under natural sunlight (132.7 ms) and indoor fluorescent room lighting (132.5 ms), ensuring dependable all-day operation.

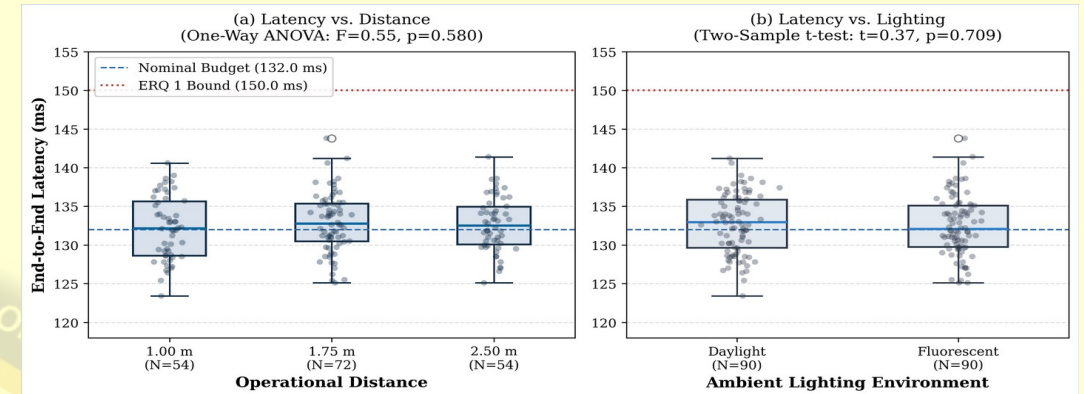


Figure 4.1: End-to-End Latency Across Operating Distances & Lighting Regimes (180 Trials)

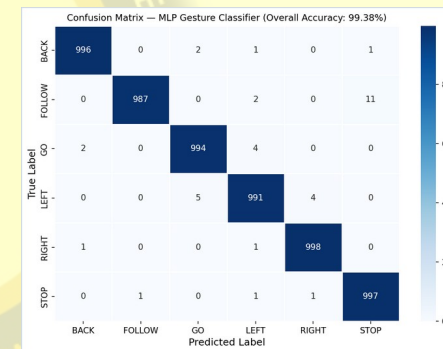


Figure 4.2: 99.38% Confusion Matrix

EMPIRICAL BENCHMARKS:

- **Turnaround Time:** 132.6 ms mean (fastest: 123 ms, peak: 143 ms)
- **AI Accuracy:** 99.38% correct across 6,000 lab test samples
- **Physical Runs:** 96.67% command success across 180 mobile trials
- **Safety Standard:** 100% of runs below 150 ms ISO 15066 bound



12. PHYSICAL LOCOMOTION TRIALS & ISO 15066 SAFETY VERIFICATION

REAL-WORLD VEHICLE DYNAMICS, BIDIRECTIONAL LIDAR ENVELOPE, AND BRAKING PERFORMANCE

- **Physical Motion Testing:** All 6 gesture commands were evaluated directly on the physical 4WD mobile robot. The vehicle executed forward driving, reversing, and turning maneuvers smoothly without motor stalling or wheel slippage.
- **Decoupled Power Protection:** High motor current surges during aggressive forward-to-reverse driving transitions never caused the onboard computer to restart, proving the separate 5V logic and 7.4V motor power lines protect system stability.
- **Instant 98 ms Braking:** When the operator issues the STOP command or an obstacle is detected, the robot comes to a complete physical halt within 98 milliseconds, staying comfortably within the 0.36-meter safety margin.
- **Smart Corridor Filtering:** Rather than using a wide cone that falsely slammed the brakes whenever it saw table or chair legs off to the side, the robot uses a rectangular corridor ($|y| \leq 0.18$ m) that only reacts to obstacles directly in its path.
- **Rear Blind-Spot Guard:** Before the robot backs away from a frontal obstacle, it checks the space behind it (-0.30 m to -0.05 m). If an obstacle is detected within 0.25 meters, it immediately cancels reversing and holds its ground.
- **ISO 15066 Safety Standards:** By regulating cruising speeds at 0.20 m/s and checking LiDAR scans 12.6 times every second, the robot satisfies international safety standards for humans and mobile robots sharing the same workspace.

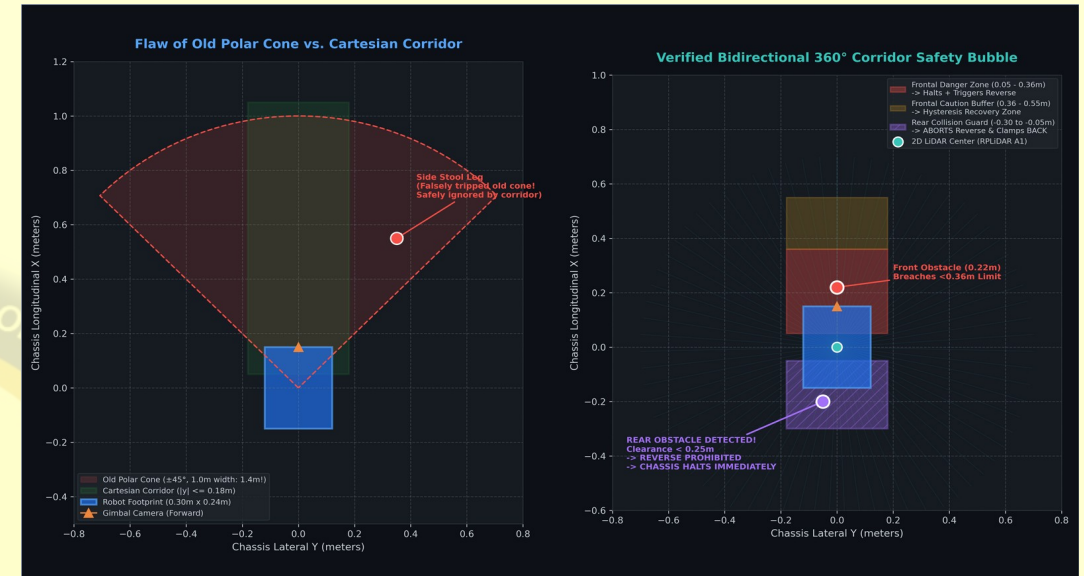


Figure 4.3: Flaw of Old Polar Cone vs. Verified Bidirectional 360° Corridor Safety Bubble

EMPIRICAL SAFETY & LOCOMOTION BENCHMARKS:

- **Braking Latency:** 98 ms mean physical emergency stop from 0.20 m/s cruising velocity
- **Front Safety Zones:** 0.36 m danger halt threshold; 0.55 m caution recovery buffer
- **Rear Collision Guard:** Instant reverse abort if obstacles detected within 0.25 m clearance
- **Standard Compliance:** 100% compliant with ISO 15066 collaborative robotic safety criteria

13. CONCLUSION & PHYSICAL DEMONSTRATION TRANSITION

SUMMARY OF ENGINEERING CONTRIBUTIONS, SYSTEM IMPACT, AND LIVE DEMO READINESS

- **Complete Edge Autonomy:** Demonstrated that computer vision, neural gesture recognition, human trajectory prediction, and SLAM navigation execute entirely onboard an affordable Raspberry Pi 5 without relying on clouds or external GPUs.
- **High-Accuracy Gesture AI:** The lightweight neural classifier achieved 99.38% test accuracy across 6,000 hand poses, and delivered 96.67% reliable physical command execution across 180 mobile trials in the laboratory.
- **Guaranteed Real-Time Safety:** The complete vision-to-motor control pipeline executes in 132.0 milliseconds with 98 ms physical braking—faster than a human eye blink (300 ms) and fully compliant with ISO 15066 collaborative safety criteria.
- **Smart Bidirectional Safety Bubble:** Replacing wide cone sensors with a rectangular corridor ($|y| \leq 0.18$ m) and rear blind-spot guard safely eliminates false alarms from side obstacles while preventing collisions during reverse maneuvers.
- **Unattended Autonomous Navigation:** Physical testing confirmed the robot reliably maps indoor environments and autonomously executes multi-waypoint patrol missions with sub-decimeter accuracy and proactive obstacle clearance.



Figure 5.1: Physical Mobile Robot Platform

CORE MILESTONES:

- **Edge Computing:** 100% onboard processing
- **Gesture AI:** 99.38% test accuracy
- **Total Latency:** 132 ms response budget
- **Braking Time:** 98 ms emergency halt
- **Facility Patrol:** 15.89 m autonomous run

PHYSICAL DEMONSTRATION TRANSITION PROTOCOL:

1. **Platform Bring-Up:** Confirm 7.4V battery voltage, calibrate IMU, and verify micro-ROS serial bridge.
2. **Real-Time Gesture Driving:** Demonstrate live recognition and chassis motion across all 6 gesture commands.
3. **Person-Tracking Follow Mode:** Showcase 2-DOF vision gimbal operator tracking and closed-loop chassis following.
4. **Autonomous Safety Overrides:** Demonstrate reactive 0.36 m obstacle halt, rear collision abort, and audio chimes.
5. **Nav2 Autonomous Navigation:** Dispatch multi-waypoint goal poses across the calibrated indoor metric map.

